

Sequence exists everywhere in our life. For example, the students scores, the cost of your shopping list, and bills of your credit cards, all of them can be described as sequences. In that, an important and common work is to sum all the numbers in a sequence. For convenience, it is quite common to show the process of such summation. However, when taking the summation, we may want to limit the range as constraints. Write a program for figuring out the sum of any sequence and show the equation of the calculation.

Input

The input contains several cases. Each case contains a range, an integer value for sequence length n , followed by a sequence of n numbers. The range can be an open interval (a,b), a closed interval [a,b], or a half-open interval [a,b), (a,b]. The input ends at * symbol.

Output

For each case, output the summation equation for all values within the given range, and the result of summation. For any negative number, you should convert it to a subtraction of a positive value rather than an addition of a negative value. If all values are out of range, output “**All values are out of range.**”

Sample Input

[-11,81) 2 -70 32
(38,95) 5 93 75 -45 -80 -20
[-29,40] 10 -50 -41 88 -2 -1 36 73 -85 35 41
(-26,32) 4 32 61 65 -45
*

Sample Output

32 = 32
93 + 75 = 168
-2 - 1 + 36 + 35 = 68
All values are out of range.